

Exp No:2 Upload and Analyse the data set given in csv format and perform data preprocessing and visualization.

AIM:

Upload and Analyse the data set given in csv format and perform data preprocessing and visualization.

Description: Use sample data set sales-data.csv.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

file_path = 'sales-data-new.csv'
df = pd.read_csv(file_path)
print(df.head())
print(df.isnull().sum())
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
print(df.describe())
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print(product_summary)
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
```

```
plt.title('Total Sales by Product')
```

```
plt.show()
```

```
df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
```

```
sales_over_time = df.groupby('Date').agg({
```

```
    'Sales': 'sum'
```

```
}).reset_index()
```

```
plt.figure(figsize=(10, 6))
```

```
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
```

```
plt.xlabel('Date')
```

```
plt.ylabel('Total Sales')
```

```
plt.title('Sales Over Time')
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

```
pivot_table = df.pivot_table(
```

```
    values='Sales',
```

```
    index='Region',
```

```
    columns='Product',
```

```
    aggfunc='sum',
```

```
    fill_value=0
```

```
)
```

```
print(pivot_table)
```

```
# Correlation matrix
```

```
correlation_matrix = df[['Sales', 'Quantity']].corr()
```

```
print(correlation_matrix)
```

```
plt.figure(figsize=(8, 6))
```

```
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
```

```
plt.title('Correlation Matrix')
```

```
plt.show()
```

OUTPUT:

	Date	Product	Sales	Quantity	Region
0	01-01-2023	Product A_New	223	5	North_New
1	02-01-2023	Product B_New	173	4	South_New
2	03-01-2023	Product A_New	243	6	North_New
3	04-01-2023	Product C_New	323	7	East_New
4	05-01-2023	Product B_New	203	5	West_New

Date 0

Product 0

Sales 0

Quantity 0

Region 0

dtype: int64

	Sales	Quantity
count	16.000000	16.000000
mean	260.500000	6.375000
std	64.031242	1.746425
min	173.000000	4.000000
25%	210.500000	5.000000
50%	248.000000	6.500000
75%	325.500000	8.000000

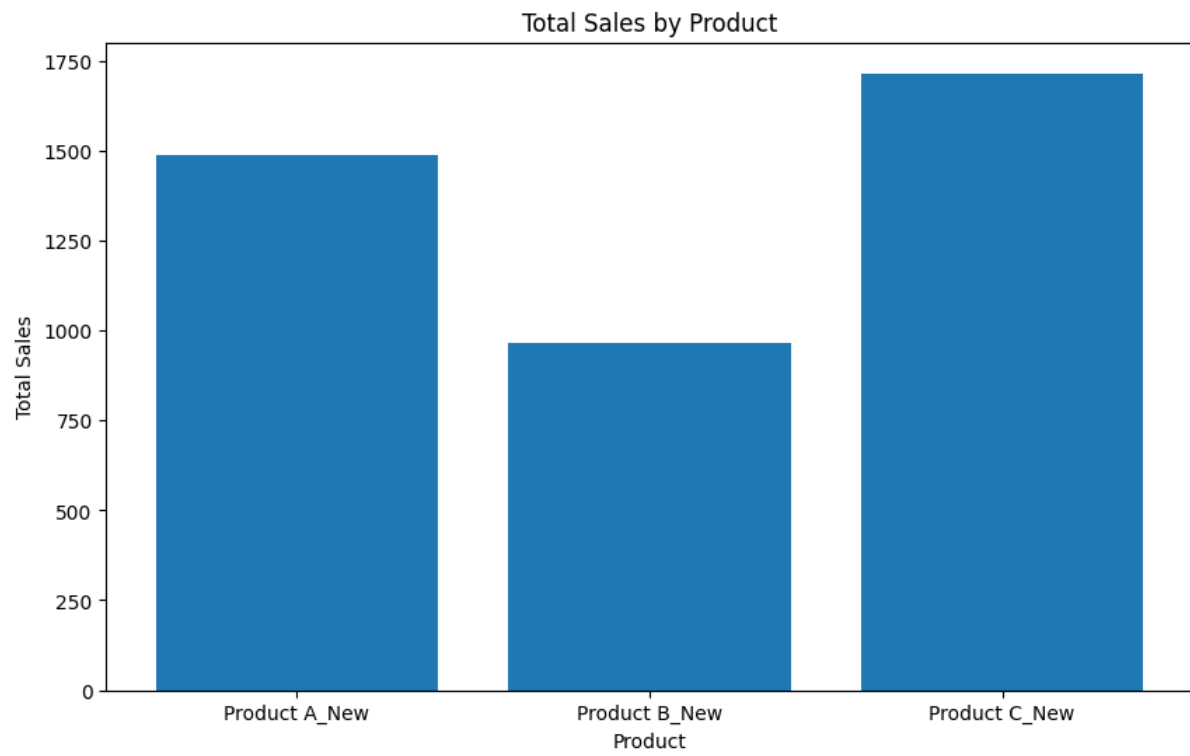
max 363.000000 9.000000

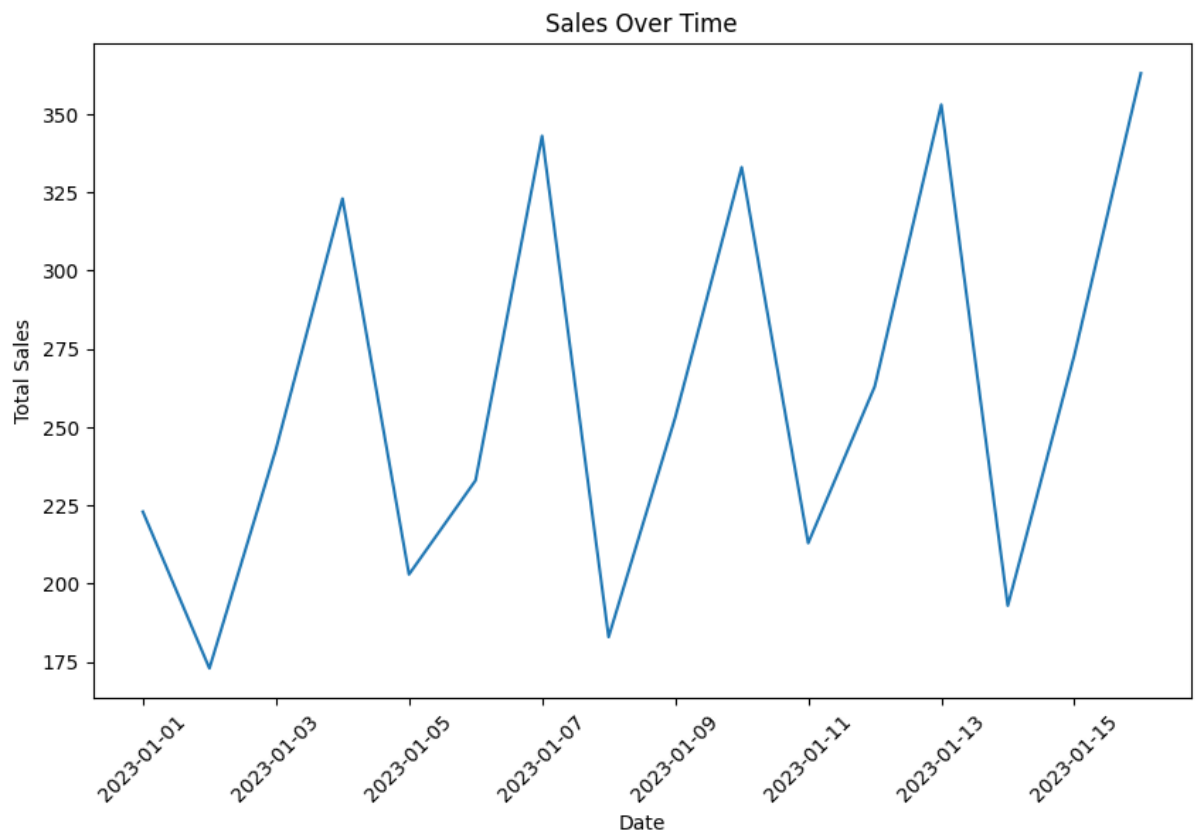
Product Sales Quantity

0 Product A_New 1488 39

1 Product B_New 965 22

2 Product C_New 1715 41





Product Product A_New Product B_New Product C_New

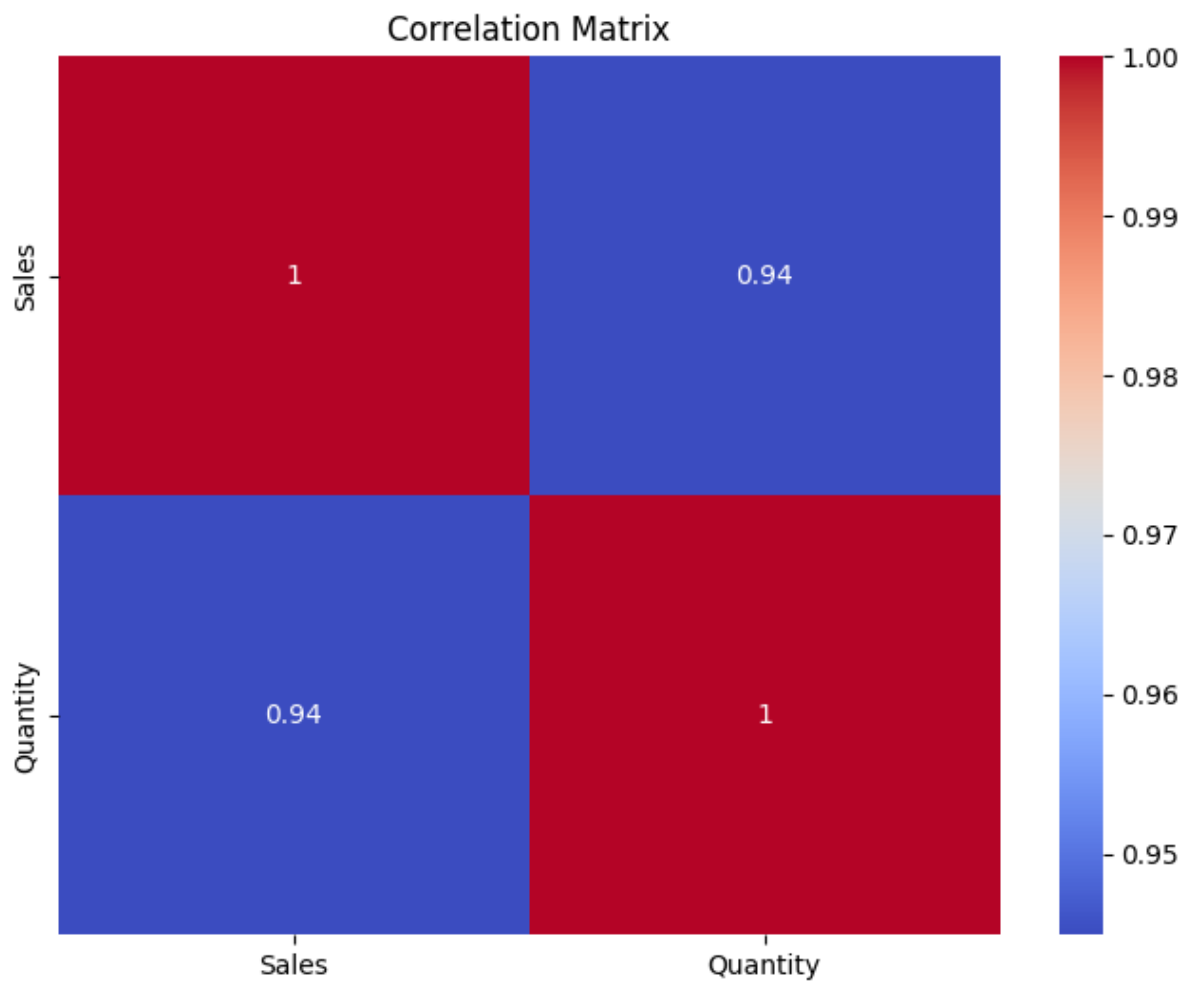
Region

East_New	0	0	1715
North_New	1488	0	0
South_New	0	549	0
West_New	0	416	0

Sales Quantity

Sales 1.000000 0.944922

Quantity 0.944922 1.000000



RESULT:

The sales data was successfully analysed and chart showed product-wise sales,sales trend,and the correlation between sales and quantity.